Bounded-Authority Agent Specialization for Adaptive Technical Interview Systems: Coordination, Safeguards, and Empirical Trade-offs

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Abstract. Adaptive technical-interview systems increasingly combine automated assessment, reinforcement learning, and language-model generation, but a collection of modules is not automatically a useful multi-agent system. PrepAIred is studied here as a centrally coordinated architecture with bounded Strategy and Coaching agents: the Strategy Agent maintains a weakness ledger and proposes difficulty or probe actions, while the Coaching Agent maintains coaching history and proposes presentation changes. A deterministic coordinator validates typed proposals, enforces authority boundaries, arbitrates against a frozen PPO policy, and commits state. Coordination was evaluated by deterministic replay of 8 recorded answer sets under 20 question-order permutations, yielding 160 sessions per condition. Relative to the frozen baseline, the full configuration changed the difficulty path in all 160 sessions and changed mean level changes by -2.67 [cluster-bootstrap 95% CI $-4.50, -0.24$] and reversals by -1.88 $[-3.12, -0.27]$. However, the split and unified controllers made identical decisions over 1,280 turns under matched rules, so decomposition itself was not shown to improve decisions. Fault injection found 38/40 adversarial-valid sessions with guardrail bypass without arbitration versus 0/40 with arbitration, while a grounding probe accepted unsupported paraphrased credit in 57/57 unambiguous cases. The evidence therefore supports bounded authority and property-specific safeguards as architectural mechanisms, not a general claim of multi-agent superiority.

Keywords: multi-agent systems · bounded authority · agent coordination · technical interview assessment · reinforcement learning · fault injection

1 Introduction and Research Questions

Technical interview preparation requires candidates to demonstrate domain knowledge, structured reasoning, problem solving, and communication under interactive and time-constrained conditions. Static question repositories provide limited adaptation, while unconstrained language-model interviewers can mix dialogue generation with assessment decisions and can drift from a fixed rubric. Automated short-answer assessment has likewise progressed from lexical matching

toward semantic representations, but semantic similarity alone does not guarantee that a response establishes the required reasoning [1,2].

Adaptive difficulty creates a second systems problem. Reinforcement-learning methods can choose actions from a state trajectory [7,10], yet a learned policy does not itself specify how persistent conceptual weaknesses, coaching history, evidence grounding, and safety constraints should be coordinated. A third problem is authority: when a language model or multiple adaptive components can both evaluate and act, the resulting workflow may lack an explicit boundary between evidence generation and state mutation.

PrepAIred addresses this systems problem with a centrally coordinated architecture containing two explicitly bounded agents. The Strategy Agent maintains a private weakness ledger and proposes difficulty or probe actions. The Coaching Agent maintains a coaching-history ledger and proposes feedback depth and emphasis. Neither agent writes committed state. A deterministic coordinator validates proposals, applies authority and version checks, arbitrates against the frozen policy, and remains the sole system of record. This design is evaluated as an engineering question rather than assuming that agent decomposition is inherently beneficial.

The paper asks four questions:

1. **RQ1 – Coordination:** Does the bounded agent layer change difficulty trajectories relative to the frozen single-controller baseline?
2. **RQ2 – Specialization:** Does a split Strategy/Coaching architecture produce different decisions from a functionally equivalent unified controller under matched observations and rules?
3. **RQ3 – Safeguards:** Which implemented validation, grounding, arbitration, threshold, and fault-containment mechanisms are necessary for the measured safety properties?
4. **RQ4 – Evaluator validity:** How closely does the frozen technical evaluator agree with the available human-reference benchmark, and what is known about length dependence?

The paper's contribution is deliberately limited to three points. First, it specifies an authority-separated agent architecture with private ledgers, typed proposals, and deterministic arbitration. Second, it reports replay and fault-injection evidence, including negative findings, that distinguishes architectural separation from measured decision benefit. Third, it reports a short evaluator-validity analysis and its unresolved length confound. The question bank also stores Bloom-style cognitive metadata, but this is descriptive and untested for efficacy; [**NEEDS EXPERIMENT**].

2 Related Work

2.1 Automated short-answer and interview assessment

Sentence-BERT established efficient semantic sentence representations for similarity tasks [1], while automated short-answer grading has combined semantic

similarity and entailment for educational responses [2]. Interview-oriented work has also used acoustic and multimodal features to model delivery and interview outcomes [4,5,6]. PrepAIred combines semantic, concept, and CrossEncoder evidence for technical-answer assessment and keeps communication confidence separate from technical correctness.

2.2 LLM-as-judge and grounded generation

Language models can generate flexible feedback but can also introduce unsupported content and evaluator inconsistency [9,13,17]. LLM-as-judge studies have identified systematic evaluation effects and motivated more controlled scoring procedures [17]. PrepAIred therefore gives the language model a downstream generation role rather than numerical scoring authority. Evidence-grounded text evaluation provides a closely related design principle for constraining generated judgments [16].

2.3 Adaptive tutoring and runtime enforcement

Cognitive tutoring and knowledge-tracing approaches model learner state for instructional sequencing [10]. PPO provides a practical policy-gradient algorithm for continuous observations and discrete actions [7]. Safe reinforcement-learning work has formalized shielding and external safety envelopes that can override learned actions [18,19]. PrepAIred similarly places deterministic guardrails and arbitration around a frozen PPO policy.

2.4 Multi-agent coordination and fault injection

Agent research distinguishes autonomous, reactive, proactive, and social behavior [12]. Modern LLM agent frameworks include conversational coordination and role-based workflows, such as AutoGen and MetaGPT [8,15]. Recent audits identify coordination collapse and specification drift in LLM multi-agent systems [3]. Fault-injection methods for LLM multi-agent systems, including the Huang et al. ICML 2025 work and MAS-FIRE, are relevant to the present safety analysis, but the exact bibliographic records are not reproduced here: **[CITATION TO VERIFY]**. The present study differs by testing a bounded-authority architecture in which agents propose but the coordinator commits state.

3 System Architecture and Bounded Authority

The deployed system consists of the interview interface, FastAPI/WebSocket gateway, orchestration hub, ASR and response-analysis components, deterministic technical evaluator, communication-confidence component, frozen PPO policy, question selector, and a local LLM generation service. The agent layer explicitly adds Strategy and Coaching roles around this existing hub.

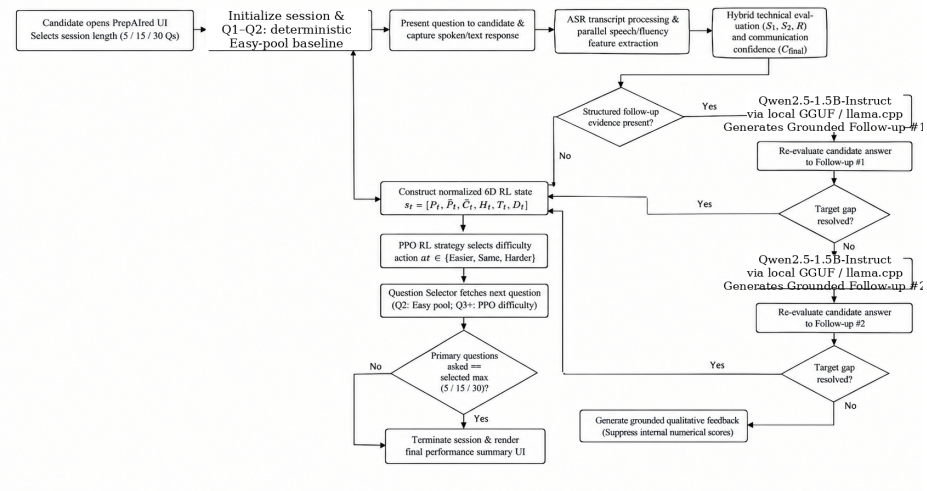


Fig. 1. Runtime workflow of PrepAIred. The candidate response is transcribed and analyzed for technical evidence and communication state; structured evidence can trigger bounded follow-up generation. The adaptive path constructs the normalized six-dimensional state and selects the next difficulty under the coordinator's arbitration and guardrail rules. The figure is a workflow view; the agent authority model is specified in Sect. 3.

3.1 Strategy Agent

The Strategy Agent maintains a private weakness ledger containing persistent conceptual deficiencies and score trajectories. It proposes Easier, Same, or Harder actions and can propose a targeted follow-up probe. It cannot overwrite evaluator scores, directly write difficulty, mutate committed session history, terminate a session, or override guardrails.

3.2 Coaching Agent

The Coaching Agent maintains a private coaching-history ledger containing previously delivered hints/examples, repeated misconceptions, and concepts to emphasize. It proposes feedback depth, emphasis, remediation tags, and presentation changes. It has no authority over numerical scores, difficulty actions, follow-up count, or session termination.

3.3 Coordinator and typed proposals

The coordinator is the sole state writer. Each observation and proposal is a typed object containing session and turn context and the current state version. Unknown fields, non-finite values, out-of-range values, stale versions, and unauthorized fields are rejected. Accepted proposals are recorded with provenance and a decision reason; rejected proposals do not mutate state.

3.4 Formal bounded-authority property

Let $\Delta(a)$ denote the state mutation produced by an agent output a. Bounded authority requires:

$$a \in \mathcal{A}_{\text{agent}} \implies \Delta(a) \cap \mathcal{S}_{\text{protected}} = \varnothing, \tag{1}$$

where protected state includes committed evaluator scores, the committed session record, and coordinator-owned control fields. The implementation test checks that protected fields remain unchanged after accepted agent proposals and after proposal rejection paths.

3.5 Bloom metadata

Question records also carry Bloom-style cognitive metadata. The operational bands used by the system start at Understand and proceed through Apply, Analyze, Evaluate, and Create. This is descriptive metadata; no experiment in the current evidence establishes that the Bloom labels improve difficulty control or learning. The system also refers to coarse Easy/Medium/Hard pools while maintaining a five-level numerical difficulty variable; the exact mapping between those representations is [**NEEDS VERIFICATION**].

4 Orchestration, Arbitration, and Validation

The implemented precedence order is deterministic. Hard constraints establish the legal state/action range. The guardrail layer is applied last and wins when a rule fires. A validated Strategy proposal can supersede the PPO action when it meets the evidence and confidence requirements. Otherwise the frozen PPO action, or the non-RL fallback when PPO is unavailable, remains the decision source.

The principal arbitration threshold is $\tau = 0.6$. A proposal must be structurally valid, current, evidence-grounded, and within the agent's authority. The final selected action is then passed through the guardrail layer. The training/simulation guardrail module and the deployed inline copy differ at documented boundary states; the synthetic grid contains 4,728 disagreements out of 39,930 states (11.8

4.1 Proposal validation

Every proposal passes four checks before arbitration: (1) strict schema validation; (2) forbidden-key scanning; (3) optimistic concurrency/version validation; and (4) concept-evidence grounding. An agent exception or timeout causes its proposal to be dropped and the default policy to proceed.

4.2 Worked-turn structure

A complete verbal turn has the following structure:

1. the hub builds an immutable observation;
2. agents receive that observation and return typed proposals;
3. proposals are validated for schema, authority, version, and evidence;
4. the coordinator arbitrates Strategy proposals against PPO and the deterministic rules;
5. the final action is passed through guardrails;
6. the hub commits the action and increments the state version.

No real log trace was included in the evidence set used here, so a concrete example is: [**EXAMPLE TO INSERT FROM LOGS**].

4.3 Decision-source table

The replay table reports per-session counts for Strategy-final decisions, guardrail fires, and PPO agreement, but it does not provide a disjoint partition of every final decision among Strategy, PPO, guardrails, and hard constraints. A defensible share breakdown therefore requires raw decision-source records: [**RESULT PENDING - NOT RUN/DERIVED**].

5 Technical Evaluator and Communication Model

5.1 Technical evaluator

Let S_1 be sentence-level semantic similarity, S_2 concept coverage, and R the normalized CrossEncoder signal. The implemented base score is:

$$S_{base} = 0.15S_1 + 0.35S_{2,eff} + 0.50R. \tag{2}$$

Concept coverage uses a 0.30 cosine threshold and is damped to $0.60S_2$ when $R \leq 0.30$:

$$S_{2,eff} = \begin{cases} S_2, & R > 0.30, \\ 0.60S_2, & R \leq 0.30. \end{cases} \tag{3}$$

Bonus matches add 0.03 at similarity above 0.50. Explicit mistake-vector matches incur 0.07 each, capped at 0.30. The available source also contains weak-evidence penalties of 0.20 and 0.05; because these terms do not appear consistently across the earlier evaluator descriptions, they are retained here only as a model item to be verified against the evaluator code: [**NEEDS VERIFICATION**]. If confirmed, the full deployed formula is:

$$S_{final} = \text{clamp}\left(S_{base} + S_{bonus} - P_{mistake} - P_{weak} - P_{hedge}, 0, 1\right), \tag{4}$$

with $P_{hedge} = 0.03$ for the documented low-coverage/low-reasoning condition. The mandatory-concept ceiling is 0.60.

The CrossEncoder output mapping is implementation-sensitive. Paper evidence notes version-dependent raw-logit versus sigmoid behavior; therefore the exact ‘sentence-transformers‘/‘transformers‘ and CrossEncoder library versions used to produce the reported score mapping must be inserted here: [**NEEDS VERIFICATION - exact library versions**]. The paper does not use the phrase “raw logit” without that verification.

5.2 Communication confidence

Communication is an independent state variable. The documented acoustic terms are:

$$C_j = \text{clamp}(1 - j/0.03), \tag{5}$$
$$C_s = \text{clamp}(1 - s/0.12), \tag{6}$$
$$C_h = \text{clamp}(h/28), \tag{7}$$
$$V_Q = 0.32C_j + 0.30C_s + 0.28C_h + 0.10C_{pitch}. \tag{8}$$

Fluency, rate, and rhythm are combined with linguistic confidence into C_{final}. The configured speaking-rate band is 2.1–4.3 words/s; the final implementation should be checked for unit consistency where the producer/consumer historically used different labels. No psychological-validity claim is made.

6 Methodology

6.1 Evidence classes

Four evidence classes are distinguished. REPLAY uses deterministic replay through the real orchestration hub. FAULT-INJECTED removes or perturbs safeguards. SOCKET checks exercise the communication boundary. SYNTHETIC probes enumerate controlled state configurations such as the guardrail grid.

Protocol tags are local and carry no external timestamp. Experiment XC was designed after XA and XB and was prospectively tagged; no external registration timestamp is associated with it.

6.2 Replay design

The coordination study uses 8 recorded answer sets and 20 question-order permutations (seeds 42–61), producing $8 \times 20 = 160$ sessions per condition. The sessions reuse the same eight structural clusters and are therefore dependent. Cluster bootstrap resamples the eight clusters rather than treating the 160 permutations as independent.

Table 1. Replay ablation summary. Counts are per condition; trajectory and latency quantities are mean per session unless stated otherwise.

Cond.	Compl.	Trans.	Chg.	Rev.	PPO agr.	Strat.	p50 ms
B0	160	160	15.119	11.744	0.626	0.000	0.546
B7	160	160	2.675	1.256	–	0.000	0.463
B0c	160	160	15.119	11.744	0.626	0.000	0.553
M1	160	160	12.450	9.869	0.400	6.669	1.589
M2	160	160	15.119	11.744	0.626	0.000	1.218
M3	160	160	12.450	9.869	0.400	6.669	1.118
M4	160	160	2.225	1.000	0.014	19.619	1.573
M5	160	160	12.450	9.869	0.400	6.669	1.581
M6	160	160	12.450	9.869	0.400	6.669	1.592
M7	160	160	2.706	1.312	–	9.037	1.502
M8	160	160	12.450	9.869	0.400	6.669	1.575

6.3 Experimental conditions

The main replay conditions are B0 (frozen baseline), B7 (PPO off, no agents), B0c (canonical guardrails), M1 (full bounded-agent configuration), M2 (silent agents), M3 (Coaching ablated), M4 (arbitration removed), M5 (grounding check removed), M6 (Strategy-to-Coaching communication removed), M7 (PPO policy disabled), and M8 (additional coordination ablation). The split-versus-unified comparison uses the specialized two-agent implementation and a monolithic controller that receives the same observations, thresholds, and arbitration rules and maintains a joint session ledger.

6.4 Statistical analysis

The replay study uses non-parametric percentile bootstrap intervals with 10,000 resamples. Cluster bootstrap resamples the eight underlying clusters. Leave-one-cluster-out analysis checks sensitivity to an individual cluster. No p-values are used for the replay comparisons. For binary containment results, exact Clopper–Pearson 95% intervals are used.

7 Results: Coordination and Specialization

7.1 XA: coordination dynamics

Every replay condition completed 160 sessions and preserved evaluator-score authority and valid difficulty transitions. The exact Clopper–Pearson lower bound for a 160/160 property is 0.977.

M1 changed the difficulty path in all 160 sessions. Compared with B0, mean level changes differed by -2.669 with a cluster-bootstrap 95% CI of $[-4.50, -0.24]$,

and reversals differed by -1.875 with CI $[-3.12, -0.27]$. The corresponding PPO-agreement difference was -0.226 with CI $[-0.284, -0.170]$.

The correct interpretation is behavioral stability, not quality improvement. B7, which disables PPO and agents, has only about 2.7 level changes per session versus 15.1 in B0 and 12.45 in M1. Therefore "fewer level changes" cannot be treated as evidence of better adaptation.

The Strategy Agent generated 3,814 proposals across the replay: 1,300 remediate proposals, 1,240 recover proposals across the recorded confidence values, and 1,274 steady proposals. It generated zero stretch proposals. The extra +0.59 probe turns per session were driven entirely by one replay cluster; the other seven clusters contributed 0.00 extra turns.

7.2 XB: fault injection

Fault-injection experiments evaluated crash, timeout, malformed, forbidden-key, stale, NaN, adversarial-valid, and ungrounded fault types. Every evaluated cell retained score authority and valid transitions in 160/160 sessions, giving a Clopper–Pearson interval of $[0.977, 1.000]$. Because containment was already at the ceiling in every cell, that global property does not discriminate among safeguards.

Schema validation alone contained forbidden-key output in the corresponding injected condition. For the adversarial-valid fault, removing arbitration produced guardrail bypass in 38/40 sessions (95%; exact 95% CI $[0.831, 0.994]$), whereas enabling arbitration produced 0/40 bypasses (0%; CI $[0.000, 0.088]$). These are sessions, not individual fault cases.

For the per-agent breaker test, the breaker opened in 29/40 sessions (72.5%; 95% CI $[0.561, 0.854]$), below the predicted $\geq 80\%$ threshold. This is Contradicted Prediction B6. The injected fault rate used in the fault-injection configuration was 0.3 (12/40; 95% CI $[0.166, 0.465]$). Pooled p95 latency was about 154 ms, close to the 150 ms timeout configuration.

For ungrounded emphasis, removing the grounding check produced the targeted failure in 40/40 sessions, while keeping the check produced 0/40. A well-formed but substantively wrong proposal remained undetected by structural validation, demonstrating that schema, authority, version, and syntax checks are not semantic correctness proofs.

7.3 Safeguard-by-property matrix

7.4 XC: threshold decomposition

Only one threshold sweep is reported. At $\tau = 0.0$, mean level changes collapsed to 2.225 per session and PPO agreement to 0.014, reproducing the qualitative M4 collapse. At $\tau = 0.5, 0.6, 0.7$, level changes remained 12.450 and PPO agreement 0.400. At $\tau = 0.8$, level changes remained 12.450 and PPO agreement was 0.404. At $\tau = 0.9$, the Strategy Agent was silenced and the trajectory returned to the B0 baseline: 15.119 changes and 0.626 PPO agreement.

Table 2. Measured safeguard properties. "Not isolated" denotes a mechanism for which the supplied evidence does not provide a single-factor fault result.

Safeguard	Score authority	Valid transition	No bypass / leak	Fault containment
Schema validation	160/160 overall	160/160 overall	forbidden-key contained	Not isolated
Authority check	160/160 overall	160/160 overall	unauthorized fields rejected	Not isolated
Version check	160/160 overall	stale proposals rejected	Not isolated	Not isolated
Grounding check	160/160 overall	160/160 overall	40/40 ungrounded failures without check; 0/40 with check	Not isolated
Confidence threshold	160/160 overall	threshold sweep preserved legal states	$\tau = 0$ reproduced trajectory collapse	Not isolated
Arbitration	160/160 overall	160/160 overall	adversarial-valid bypass: 38/40 without vs 0/40 with	supports fail-closed path
Guardrails	160/160 overall	160/160 overall	M4 bypass observed	Not isolated
Per-agent breaker	160/160 overall	160/160 overall	Not a semantic validator	29/40 opened under injected fault

Table 3. Contradicted predictions retained in the merged analysis.

ID	Prediction	Result
A9	PPO unavailable would change at least 50% of session paths	4/160 sessions (2.5%); contradicted
A10	Canonical guardrails would change at least one path vs. legacy	0/160; contradicted
B6	Per-agent breaker would open in at least 80% of injected sessions	29/40 (72.5%); contradicted

The realised Strategy confidence values were only 0.45, 0.70, and 0.80. The plateau between 0.5 and 0.8 is therefore a consequence of the observed discrete confidence values, not evidence of robustness to continuous threshold variation.

7.5 XD: grounding probe

The lexical grounding check accepted unsupported paraphrased credit in 57/57 unambiguous cases. Seventeen earlier apparent rejections were attributable to artifacts of the check. The grounding result therefore constrains interpretation of the safeguard: the implemented check can enforce a specific lexical/evidence contract, but it is not established as a semantic grounding verifier.

7.6 Contradicted predictions

8 Specialization Consistency and Overhead

A turn is one replayed candidate-response/evaluator-state update. The matched replay contains 64 recorded answer instances under 20 question-order permutations, giving $64 \times 20 = 1,280$ turn-level observations, grouped into 160 sessions across the eight underlying clusters.

The unified controller, 'UnifiedCoachAgent', maintains one joint ledger and synthesizes strategy, follow-up, and coaching outputs from the same observation, thresholds, and arbitration rules given to the split architecture. It produced identical decisions and difficulty trajectories across all 1,280 turns. This result is expected when two implementations execute the same deterministic rules under matched inputs. It is therefore a consistency check, not evidence that the specialized architecture is superior or necessary.

The measured costs of specialization are also explicit. M1 median pooled per-turn latency is 1.589 ms versus 0.546 ms for B0, a difference of +1.043 ms with a reported interval of approximately $[+1.00, +1.08]$ ms. The split-vs-unified component dispatch comparison is 0.0440 ms versus 0.0401 ms, and the two-model serialization micro-benchmark is 0.0092 ms versus 0.0038 ms.

9 Evaluator Validity

The evaluator study is deliberately short. It uses 64 constructed technical answers across eight questions with an available human-reference score. The CrossEncoder-only signal has Spearman $\rho = 0.4832$. The deployed composite has $\rho = 0.3812$. A simple word-count baseline has $\rho = 0.4897$. The CrossEncoder is also associated with answer length; the reported length-controlled partial correlation is 0.2434 with 95% CI $[-0.008, 0.507]$.

The paired contrast between the CrossEncoder-only and composite correlations includes zero. Therefore the benchmark does not establish that the composite is superior to the CrossEncoder-only signal, nor that the CrossEncoder adds a length-independent reasoning contribution. Per-question correlations were not reported in the supplied evidence and are therefore not added here.

10 Design Lessons

Four design lessons follow directly from the measured results.

1. **Architectural separation does not imply decision benefit.** Split and unified controllers were identical under matched deterministic rules.
2. **Thresholds need informative confidence variation.** A two-valued or three-valued realised confidence signal can make a broad threshold sweep appear flat without demonstrating robustness.

Table 4. Fault model used in the fault-injection analysis.

Fault	Scope
Crash / exception	Agent terminates unexpectedly during a turn
Timeout	Agent exceeds the 150 ms defensive timeout
Malformed proposal	Schema-invalid, NaN, or out-of-range output
Forbidden key	Proposal attempts to write protected state
Stale proposal	Proposal carries an obsolete state/session version
Ungrounded concept	Proposal cites unsupported concept evidence
Adversarial-valid	Proposal is syntactically valid and authorized but intentionally semantically wrong
Compromised hub	**Out of scope**: malicious coordinator code
Malicious agent code	**Out of scope**: compromise below the proposal interface

3. **Containment needs property-specific metrics.** Global 160/160 containment at ceiling does not tell which safeguard matters; adversarial-valid bypass and grounding-emphasis tests were more discriminating.
4. **A lexical grounding check is not a semantic verifier.** The 57/57 unsupported-paraphrase result shows the distinction directly.
5. **Lower trajectory volatility is not a quality metric.** B7 is much more stable than B0 while disabling the learned and agentic adaptation mechanism.

11 Threat Model

12 Reproducibility and Evidence Provenance

The replay question-order seeds are 42–61 and the bootstrap seed is 42 where explicitly documented. The study uses deterministic REPLAY and fixed recorded inputs for the coordination experiments. The paper's experiment tags are local and have no external timestamp. The retained replay payloads contain no names or direct identifiers according to the study description.

The exact sentence-transformers/transformers/CrossEncoder library version used for the evaluator scores remains [**NEEDS VERIFICATION**]. This is material because the evidence indicates version-dependent raw-logit versus sigmoid behavior that changes the implemented R mapping. A final reproducibility package should release anonymized replay inputs, experiment scripts, configuration hashes, and the exact software-version manifest.

13 Limitations

All coordination experiments use deterministic replay; no live generative LLM is in the loop during the replay study. Candidate learning, pedagogical benefit, interview-placement success, and live LLM hallucination rates are therefore outside the present evidence. The eight replay clusters are reused across twenty permutations, leaving eight independent structural units. The proactive +0.59-turn effect is driven by one cluster. The Strategy Agent never produced a Harder

proposal, so that branch is unexercised. The human-reference evaluator study uses constructed answers and incomplete rater provenance; its results are exploratory. The confidence model is not human-validated. A semantic grounding failure remains. The exact evaluator-library versions and the weak-evidence penalty implementation require code verification. These limitations constrain the claims to architecture, replay behavior, and tested safeguard properties.

14 Conclusion

PrepAIred demonstrates a centrally coordinated bounded-authority design in which Strategy and Coaching maintain separate ledgers and produce typed, validated proposals while the coordinator remains the sole state writer. The empirical findings are deliberately mixed. The full configuration changes replayed difficulty trajectories relative to B0, but a functionally equivalent unified controller makes the same decisions over 1,280 matched turns. Thus agent decomposition itself is not shown to improve decision quality. The stronger empirical contribution is property-specific: arbitration prevents the tested adversarial-valid guardrail bypasses, grounding checks prevent the tested ungrounded-emphasis faults, and fault handling preserves the measured state invariants. At the same time, 57/57 unsupported paraphrased grounding probes are accepted, showing a clear weakness of the current grounding method, and well-formed but semantically wrong proposals remain possible. Future work should test stronger baselines, closed-loop candidate dynamics, stronger grounding, live LLM faults, evaluator weight sensitivity, and a small human pilot under appropriate ethics review. The six future-work protocols are specified separately; until they are executed, any corresponding evidence remains [RESULT PENDING - NOT RUN].

Disclosure of Interests

The authors have no competing interests to declare relevant to this work.

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